

JAIST — Taniike Lab

Integrating OCM Literature Data into Lab-Scale ML Models

Domain-Adaptive Transfer Learning for Catalyst Yield Prediction

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Two Very Different Datasets

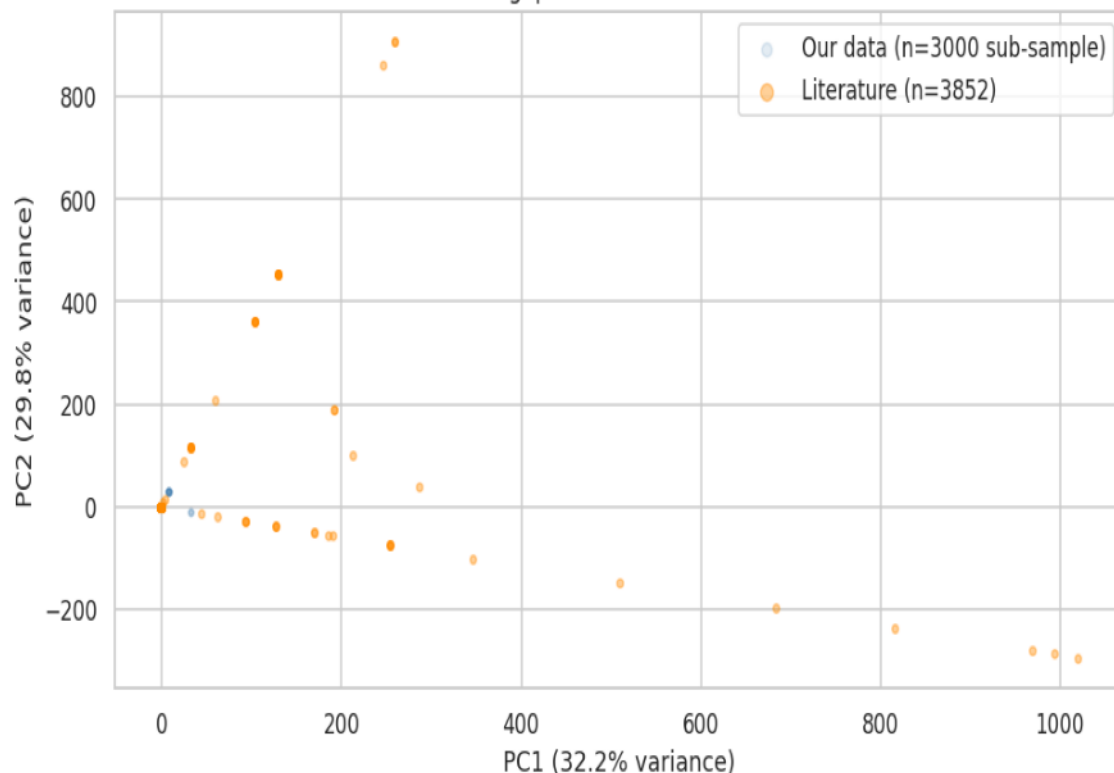
Our Lab Data (Internal)	
Samples	89,074
Year	2025
Preparation	Impregnation only
Mean Y(C2)	5.25%
✓ High volume, consistent protocol	
✗ Narrow chemical diversity	

Literature Data (Public, ≤2019)	
Samples	3,852
Year	1982 - 2019
Preparation	15+ methods
Mean Y(C2)	8.67%
✓ Chemical diversity, high-yield recipes	
✗ Publication bias, missing conditions	

Challenge: Δ mean Y(C2) = 8.67 – 5.25 = 3.42 pp ($t = -32.2$, $p = 2.3 \times 10^{-202}$) — datasets are statistically different

Domain Gap — Why We Cannot Just Mix the Data

PCA — Domain gap between our data and literature



What the PCA shows

→ 65 element features compressed to 2D

→ Blue = our 89k; Red = literature 3.8k

→ Literature spans regions our model has never seen — 78.5% are out-of-distribution

Element emphasis (not disjoint chemistry)

- Our data: Ba, Cs, Mo dominant
- Literature: Si, Na, Mn dominant
- Shared: La, Ce, Sr, Mg, Li, Ti, Ca

Naive concatenation adds noise in unfamiliar chemical regions — confirmed: RMSE worsens +5.4%

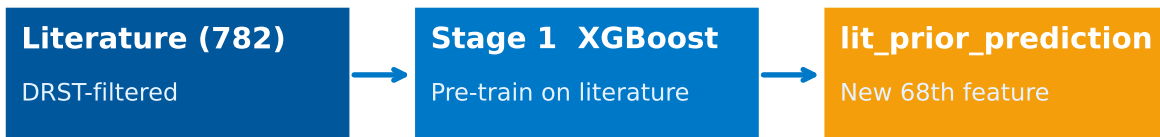
Five Approaches — What We Tried and What Happened

Method	Core Idea	CV RMSE	Δ Baseline	Result
Baseline (no lit.)	LightGBM on 89k internal samples only	2.133%	—	Reference
Naive concatenation	Add all 3,852 literature rows equally	2.248%	+5.4% ✗	WORSE
DRST ($\tau=0.30$)	Keep 782/3852 similar samples via classifier	2.019%	−5.3%	Better
KMM reweighting	QP solver: soft per-sample weights	2.035%	−4.6%	Better
Bias correction	Quantile-normalise literature Y(C2)	2.044%	−4.2%	Better
Two-stage fine-tuning ★	XGBoost prior → 68th feature → LightGBM on our labels	1.907%	−10.6% ✓	BEST

OOD test (2,139 held-out non-Impregnation samples, never trained on): Baseline 6.53% → Two-stage 3.60% = −45%

Two-Stage Fine-Tuning — How It Works

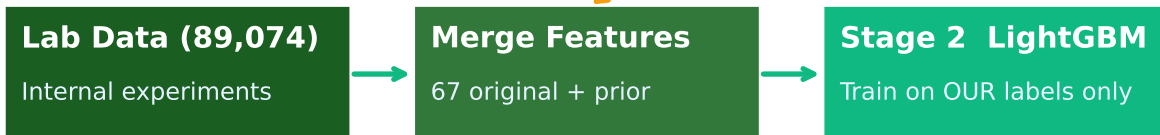
STAGE 1 — Pre-train on Literature Chemistry



Key Property

- Stage 2 trains only on our labels
- Literature bias cannot corrupt final predictions
- Prior feature = chemistry signal, not competing label

STAGE 2 — Fine-tune on Our Labels

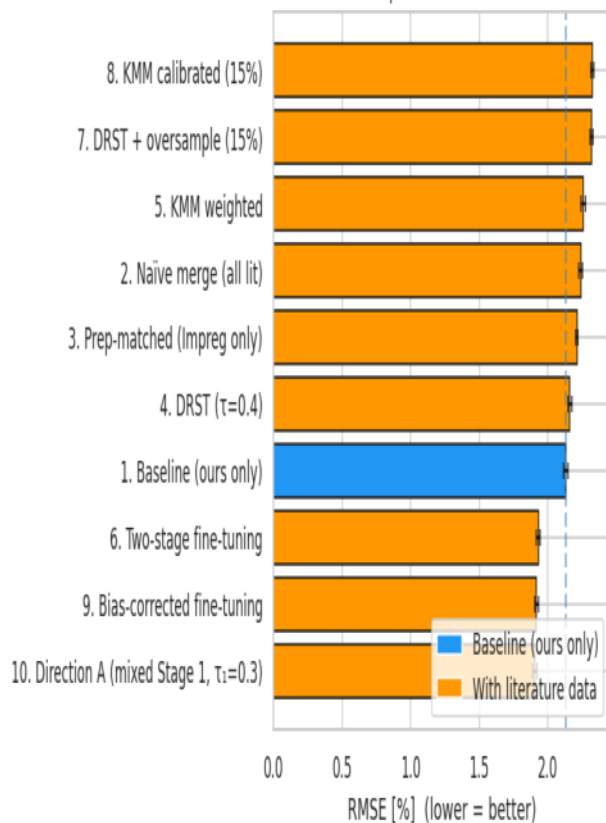


- Model learns to trust/discount the prior per composition region

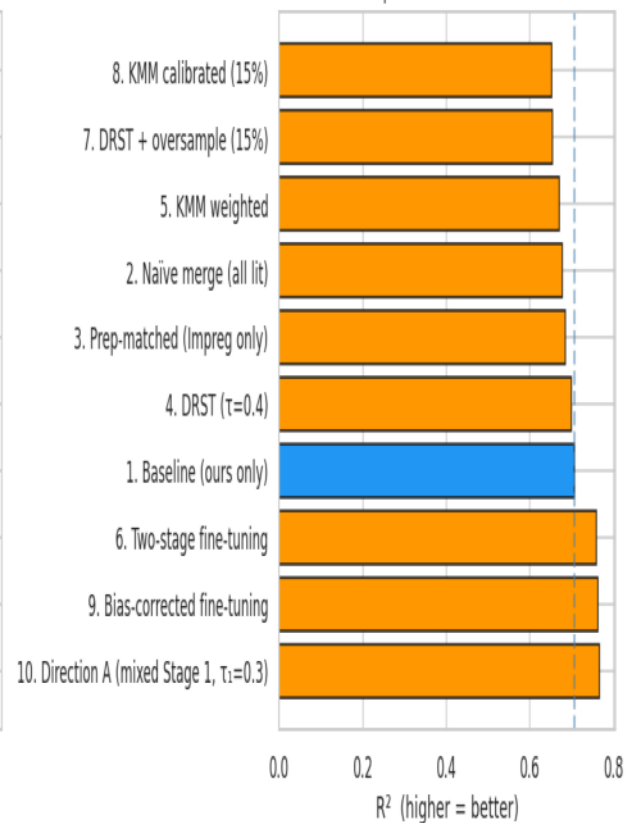
Final output

Results at a Glance

RMSE comparison — CV on our data



R^2 comparison — CV on our data



In-distribution RMSE

2.133% → 1.907%

–10.6%

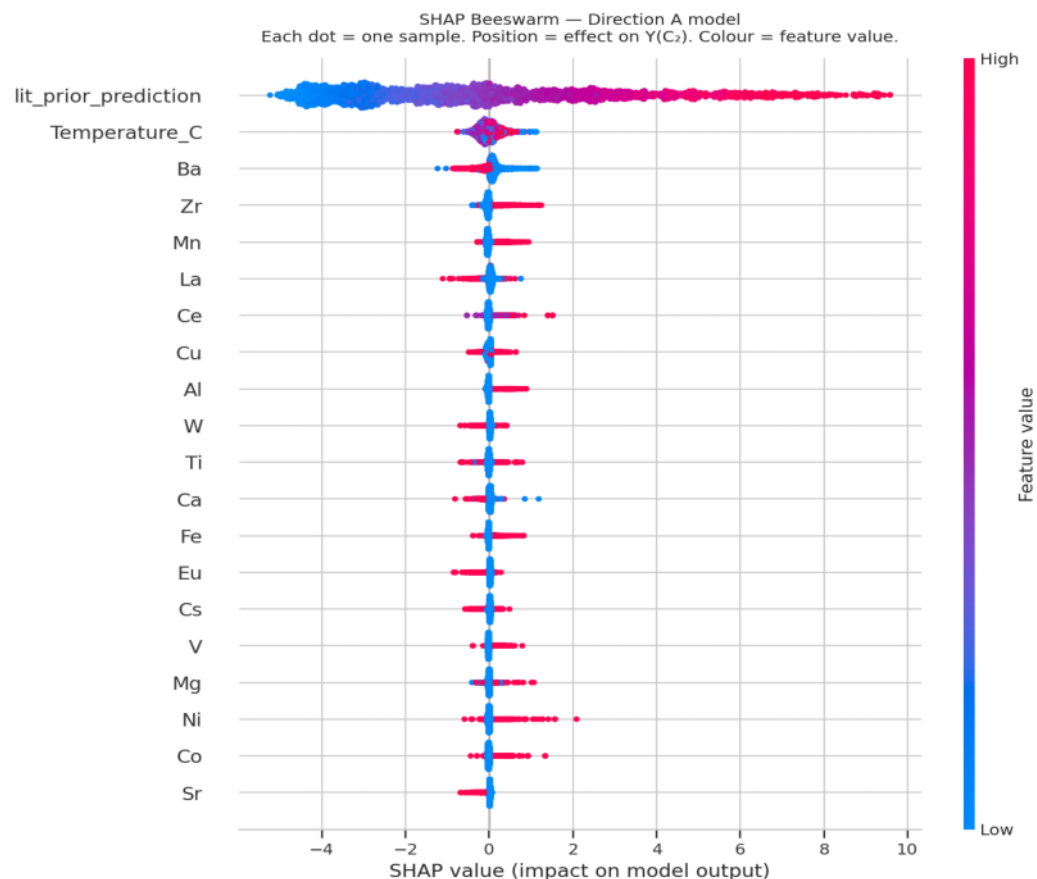
Out-of-distribution RMSE

6.53% → 3.60%

–45%

OOD test: 2,139 held-out samples, non-Impregnation methods, never seen during any training fold

What Drives the Model? — SHAP Analysis



Top findings (3,000 sample subsample)

→ #1 lit_prior_prediction — literature knowledge used on every prediction

→ Temperature — strong positive effect

→ Ba, Mn, La, Ce — all raise predicted yield

→ Li, K — mixed / context-dependent

Ceiling effect — residual by Y(C₂) range

0 - 6% RMSE 1.43-1.48% ≈ 0% bias

6 - 10% RMSE 1.95% -0.3%

10 - 15% RMSE 2.58% -1.5%

> 15% RMSE 4.70% -4.2% bias

Root cause: GHSV, CH₄/O₂, pressure absent from dataset

Limitations & Next Steps

Option A — Easy

Add Reaction Conditions

Collect GHSV, CH₄/O₂, and pressure as features. Expected RMSE drop: 4.70% → ~2.5% for high-yield range.

Option B — Medium

Tune Transfer Threshold

Grid-search DRST τ via nested CV. Current $\tau=0.30$ was manually selected. Expect ~0.3-0.5% RMSE gain.

Option C — Hard

Neural Domain Adaptation

DANN / CORAL architecture. High complexity — only justified after reaction conditions added.

Key Limitations

Missing GHSV / CH₄:O₂ / pressure • Publication bias • OOD coverage uneven

Recommended Priority

1. Add GHSV + CH₄/O₂ (highest impact)
2. Re-run two-stage
3. Tune threshold